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Microplastic contamination and risk assessment in the biotic and abiotic components of freshwater Manasbal Lake, western Himalaya, India

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Abstract
Microplastics (MPs) have emerged as ubiquitous and persistent pollutants that impact all compartments of urban and mountain environments. However, studies combining abiotic and biotic components to assess MPs abundance and characterisation from remote mountain catchments of the Himalaya are still limited. This study examined abundance, types, and ecological risks of MPs in abiotic (surface water, water column, snow) and biotic (fish) components of Manasbal Lake, an ecological hotspot in the Kashmir Valley, western Himalaya. MPs were extracted using density separation and subsequently examined microscopically to determine their abundance, shape, color, and size. The concentration ranges were observed at 59–188 MPs/L in surface water, 8–15 MPs/L in water column, 11–18 MPs/L in snow, and 7–13 MPs/individual in fish samples. The MPs shape, color and polymer composition of biotic and abiotic components were dominated by fibers, transparent color and polyethylene terephthalate (PET) type, respectively. Most of the MPs observed were in the size range of 0.1–1 mm, with significant contributions from local pollution sources. We observed higher abundances of MPs away from stream inlets or residential areas, suggesting complex hydrodynamics and wind-driven surface transport in the lake. Risk assessment indices reveal that fibers exhibit the highest risk levels, posing a significant risk to the lake ecosystem. Further, a comparative analysis with other regional western Himalayan Lakes highlights the dominance of fiber MPs, emphasizing the need for multidisciplinary research efforts to address the growing concern for MPs contamination. This study underlines the necessity for integrated catchment-scale land-use planning while advancing our understanding of MPs pollution dynamics in lake ecosystems of the western Himalaya.

Keywords Plastic, Himalaya, Lake, Snow, Microplastics, Risk assessment, Environmental sciences, Water pollution

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Introduction

Global plastic production has increased from ~1.7 million tons in 1950 to ~415 million tons in 2023, with 90% of production generated from fossil-based sources (PlasticsEurope 2024; Pastorino et al. 2023). Approximately, 2–5% of MPs find their way to oceans and lakes (Mubin et al. 2024; Yang et al. 2021). MPs are now found across all components of the earth, including biotic and abiotic matrices. They have been reported in biotic matrices such as fish, birds, mussels, crabs, and corals (Rather et al. 2025; Pastorino et al. 2021; Halstead et al. 2018; Jabeen et al. 2017; Valsan et al. 2024a; John et al. 2022; Huang et al. 2021). Similarly, MPs occur in abiotic components such as sediments, soils, water and ice/glaciers (Valsan et al. 2024b; Islam et al. 2023; Pastorino et al. 2023; Neelavannan et al. 2022; Jia et al. 2024; Shirazi et al. 2023; Warrier et al. 2022; Egessa et al. 2020; Cha et al. 2023; Sangkham et al. 2023; Jones-Williams et al. 2025; Parolini et al. 2021a, b). In addition to environmental contamination, MPs have infiltrated consumer products, including cosmetics (Bikiaris et al. 2024) and milk (Adjama et al. 2024), raising concerns about their eventual ingestion and accumulation in the human body (Lee et al. 2024). Recently, considerable research efforts have been directed towards understanding MPs abundance, bioaccumulation potential, and mobility in aquatic environments (Debnath et al. 2024; Ren et al. 2024; Sadia et al. 2024; Jessieleena and Nambi 2023; Pastorino et al. 2022).

The main pathways delivering MPs to global aquatic systems are identified as anthropogenic sources/urban runoff, sludge/biosolids, and atmospheric deposition (Kumar et al. 2025; Talukdar et al. 2023; Nash et al. 2023; Grbić et al. 2020). Public concerns for urban runoff fluxes have grown exponentially as they act as a major conduit of MPs transfer (0 to 8580 particles/L) to aquatic environments (Wang et al. 2022). Yet, recent studies have shown that MPs transported by natural and artificial vectors contaminate remote environmental systems (Neelavannan et al. 2022; Zhang et al. 2020; Napper et al. 2020). The high-mountain lakes (>1500 m), despite being less impacted and apparently pristine, show MPs contamination of biotic and abiotic components caused by point (urban discharge, tourism) and non-point (atmospheric) sources (Pastorino et al. 2023; 2022; 2021). For example, owing to low density, atmospheric long-range transport is considered a non-negligible pathway of MPs deposition in high-mountain aquatic ecosystems (Napper et al. 2020; Allen et al. 2019; Pastorino et al. 2022). Further, if ingested by aquatic/terrestrial organisms and humans, MPs can cause detrimental effects and exert localized particle toxicity. Hence, assessing the abundance and characteristic type of MPs in high-altitude lake ecosystems helps identify sources and assess human impacts in these environments (Napper et al. 2020).

Several studies have been reported from the mid and low altitude/coastal lakes of India with significant MPs contamination attributed to urban runoff, fishing, and tourism (Resmi and Vaishnavi 2025 and references therein). However, there is a lack of understanding regarding the environmental risks of MPs in the freshwater lakes of the Indian Himalaya (Tsering et al. 2022). A recent study observed a knowledge gap regarding the fate of MPs in the freshwater ecosystems and their control mechanisms in the Indian Himalayas (Talukdar et al. 2023). Another research review (Singh et al. 2024) highlighted that vast areas in western and central India, including the Himalaya, have limited reports with respect to MPs pollution. This has garnered significant attention, with a growing body of research exploring the cumulative impact of MPs in high-altitude lakes of the Indian Himalaya. Collectively, studies from the Indian Himalayan region reveal widespread MPs contamination driven by anthropogenic activities, poor waste management, and long-range transport (Ankit et al. 2024). These findings underscore the urgent need for integrated approaches to characterize MPs pollution across biotic and abiotic components, to better understand and manage local and transboundary fluxes (Tsering et al. 2022).

Despite the increasing recognition of MPs pollution globally, comprehensive assessments combining biotic and abiotic matrices from high-altitude Indian Himalayan lakes remain scarce. No study to date has evaluated both aquatic and snow compartments in conjunction with biological indicators. The present study assesses the abundance, types, contamination sources, and ecological risk of MPs in the abiotic (surface water, water column, snow) and biotic (fish) components of deep, freshwater Manasbal Lake located in the western Himalaya. The specific objectives were to: (i) quantify the abundance and distribution of MPs across surface water, water column, snow, and fish samples; (ii) evaluate their polymer types and potential ecological risks, and (iii) provide baseline data for future MPs monitoring in western Himalayan freshwater ecosystems. This study contributes critical information on the characteristics and ecological risk assessment of MPs in freshwater systems of the Indian Himalaya.

Materials and methods

Study area and environmental setting

This study was conducted at Manasbal Lake (34.25°N and 74.67°E; ~1600 m above sea level) in the Ganderbal district of Jammu and Kashmir, western Himalaya, India (Fig. 1). Recognised as the deepest freshwater lake in the Kashmir Valley (maximum depth ~13 m), Manasbal Lake is primarily fed by seasonal surface runoff, groundwater inflow, and monsoon precipitation (Lone et al. 2017). The region's precipitation comes from both the Indian

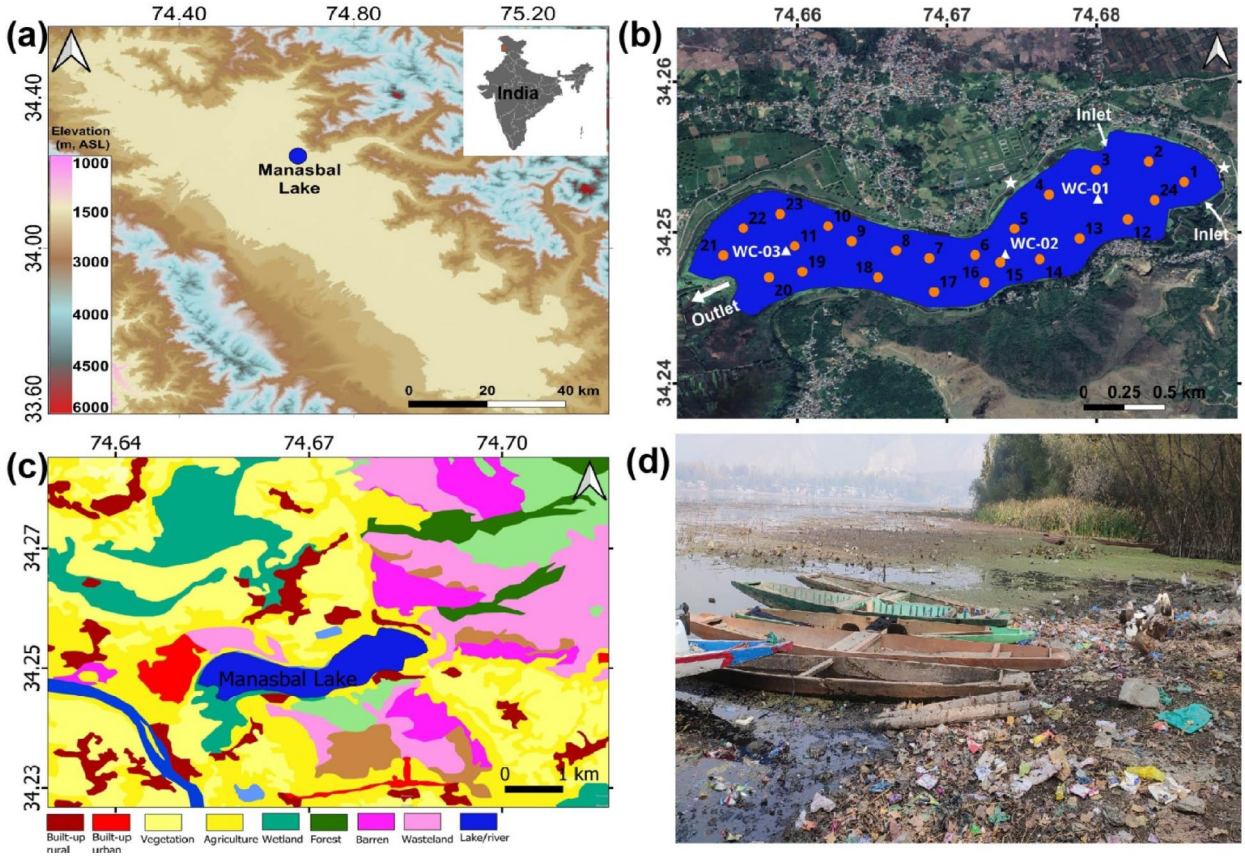

Fig. 1 (**a**) The study area map showing the location of the Manasbal Lake with elevation (**b**) the sampling stations for surface (filled circles), water column (filled white triangles) and snow samples (white stars) (**c**) the land cover map of the Manasbal Lake catchment (**d**) plastic waste and sewage dumping along the shoreline of lake

Summer Monsoon (ISM), which contributes 43% of the annual total (~440 mm) in summer, and the mid-latitude Westerlies (western disturbances) in winter and spring (Lone et al. 2020). Summer temperatures range from 10 to 20 °C, while winters are colder, ranging from ~1.5 to 10 °C. The catchment land cover map (source Bhuvan; www.bhuvan.nrsc.gov.in) includes agriculture fields, orchard plantations, urban built-up lands and parks, and pastures (Lone et al. 2017). The land cover map was generated to characterize the surrounding catchment and assess how different land-use types may influence MPs distribution in the lake.

The lake supports a diverse biota, including zooplankton (*Acanthodiaptomus denticornis*, *Diaphanosoma brachyurum*, *Ceriodaphnia pulchella*, *Alonella sp*, *Pleuroxus sp*, *Keratella quadrata*, *K. valga*, *Brachionus quadridentatus*, *Polyarthra sp.*), benthos (*Tubifex* sp.), and fish species (*Schizothorax niger*, *S. esocinus*, *Cyprinus carpio specularis*, *C. carpio communis*, *Neomacheilus latius*) (Zutshi et al. 1980). The lake catchment comprises urban built-up areas, agricultural land, horticultural vegetation, and exposed rock surfaces.

Manasbal Lake serves as a crucial freshwater resource, providing drinking water, irrigation, and supporting tourism activities such as *shikara* (boat) rides. However, runoff from residential and agricultural areas introduces nutrients, trace metals, MPs, and sewage, contributing to eutrophication (Lone et al. 2017). Its combination of seasonal inflows and anthropogenic pressures makes it a suitable representative of western Himalayan freshwater systems. As per the 2011 census, the district population was ~297,000, with local livelihoods centred on fishing, animal husbandry, farming, and seasonal tourism.

Sampling strategy

Water samples

Surface lake water, water column, and fish samples were collected in October 2022 to quantify and characterise MPs contamination in both biotic and abiotic components (Fig. 1b). Twenty-four lake surface water samples were collected from evenly distributed sampling sites (Fig. 1b). At each site, 1-L of bulk surface water was collected and sieved in situ through a 0.1 mm stainless-steel mesh, and retained material was transferred to pre-cleaned stainless-steel containers. Water column samples

were collected from three sites (WC-01 to WC-03) at three depths-surface (S), middle (M), and bottom (B)-to a maximum depth of 8 m at WC-02 (Fig. 1b). At each depth, 5 L of water were collected using a Niskin horizontal water sampler (General Oceanics), ensuring the sampler remained above the lake bed to avoid sediment disturbance. The larger sample volume (5 L) reflects the Niskin sampler's capacity and ensures adequate recovery from deeper layers. Surface samples were standardized for cross-comparison; column samples were integrated across depths to capture vertical variability. The samples were sieved and stored as per surface water protocols.

Snow samples

Eight snow samples were collected from catchment surfaces in January 2023, following a sustained snowfall event induced by a western disturbance. Sampling sites, located on opposite sides of the lake (Fig. 1b), were free of human disturbance and visible contamination. Using pre-cleaned stainless-steel scoops, fresh snow to a depth of 5 cm was collected in four replicates per site, stored in pre-rinsed 2-L metallic containers, thawed at room temperature, measured, and sieved through a 0.1 mm mesh (Ohno and Iizuka 2023; Pastorino et al. 2021; Villanova-Solano et al. 2023; Abbasi et al. 2022).

We selected the 0.1 mm mesh to reliably capture MPs fractions within the analytical resolution limits of ATR–FTIR. While smaller MPs and nanoplastics may pose higher ecological risks, particles < 100 μm are prone to analytical artefacts using conventional ATR (Attenuated Total Reflectance) - FTIR (Fourier Transform Infrared Spectroscopy; Circelli et al. 2024; Käppler et al. 2016).

Fish samples

Nine individuals of *Cyprinus carpio* were purchased from local fishermen, selected due to their abundance and consumption. Fish samples were measured (length, weight), dissected using stainless-steel tools, and the gastrointestinal tracts were removed, weighed, and wrapped in aluminium foil. Biological matter was digested with potassium hydroxide (10%) for 2–3 days (Koongolla et al. 2020; Giani et al. 2019); if residues persisted, 30% of hydrogen peroxide was added. Clear solutions were sieved using the water sample protocol.

Laboratory analysis of MPs

The MPs extraction followed standard protocols (Valsan et al. 2023; Warrier et al. 2022; Rodrigues et al. 2018a, b). Samples were sieved through a 5 mm mesh to remove large particles (> 5 mm). Further, the collected residue was transferred to a glass beaker and oven-dried at 50 °C. After drying, the residue was treated with 30% hydrogen peroxide to eliminate any organic constituents and recover particles. Later, density separation was performed on the organic-free residue using zinc chloride ($ZnCl_2$) solution with a density of 1.6–1.7 g/cm^3. Finally, the supernatant fraction containing MPs was passed through a stack of sieves (1 mm, 0.3 mm, and 0.1 mm) and collected in separate watch glasses for complete dryness at 50 °C.

MPs were quantified and photographed using a stereozoom microscope (Nikon SMZ; 40× magnification) and classified by size, shape (fibre, film, pellet, foam, fragment), and colour. The hot needle test was performed on selected particles of uncertain nature to avoid any misidentification (Marine and Environmental Research Institute (MERI) 2015). Further, for the polymer composition of the extracted MPs, 50 representative samples (0.3–1 mm) covering all shapes and colours were determined using FTIR with ATR. The selection was based on the frequent occurrence of visually similar particle types across the different matrices (surface water, snow, water column, and fish), while ensuring that all shapes and colors were adequately represented. In a few instances, although the spectral correlation indicated a positive match, the confidence score fell below the acceptable threshold for reliable identification; these particles were therefore categorised as "others". The analysis was performed using a Shimadzu IRSpirit spectrometer fitted with a QATR-S single-reflection ATR accessory, which allows non-destructive identification of polymer types through their infrared absorption characteristics. Before measurement, each particle was carefully cleaned and air-dried to eliminate any surface contaminants that might affect spectral clarity. Spectra were recorded in the 4000–400 cm^{-1} range, with 20 scans per sample to achieve high-quality signal resolution. The polymer composition was determined by comparison with reference spectra. We used the open-source spectral library "Open Specy" tool (Cowger et al. 2020) to identify and validate the polymer composition (Pearson correlation coefficient > 0.7), following which each spectrum was subjected to manual verification of key characteristic peaks, including fingerprint regions and functional group assignments, to confirm polymer identity. Spectra with ambiguous or weak matching profiles were excluded from analysis.

Contamination control (QA/QC)

Proper care was taken at every stage of the sample collection and analysis to avoid contamination. The sampling and analyses were conducted following the latest quality assurance (QA) and quality control (QC) standards (Prata et al. 2021; Brander et al. 2020). All field and laboratory apparatus employed in the present study were made of glass and stainless steel and thoroughly rinsed with Milli-Q water before use. Cotton laboratory coats were worn during each experiment to avoid contamination from synthetic textiles. The samples were always

covered with aluminium foils to avoid fallout from the airborne particles (Liu et al. 2021). All solutions and double-distilled water were filtered through a 0.1-mm sieve to ensure no prior contamination. Further, procedural blanks were performed to ensure that no in-situ MPs are transferred to the samples during experiments. The blank samples were processed similarly to the actual samples and checked for any intrinsic MPs. Procedural blanks showed negligible contamination (< 1 MPs per blank).

Ecological risk assessment

We calculated the risk assessment factor to account for MPs pollution and polymer toxicity for each component of the lake. Polymer hazard index (PHI), potential ecological risk index (PERI), and coefficient of microplastic impact (CMPI) were the risk appraisal indices used in this study (Islam et al. 2023; Pandey et al. 2023; Shirazi et al. 2023; Rangel-Buitrago et al. 2021). The Polymer Hazard Index was determined using the equation:

PHI = ∑Sn x Pn; where Sn and Pn represent the proportion of polymer types and the hazard score of the polymers, respectively (Lithner et al. 2011). This index prioritizes polymers of greater environmental and ecological concern with five hazard levels (I = 0–1, II = 1–10, III = 10–100, IV = 100–1000, and V > 1000).

The PERI is used to evaluate the potential risk to the ecology of a given environmental system and is calculated as:

$$C_f^i = \frac{C^i}{C_n^i}$$

$$T_r^i = \sum_{n=1}^{n} \frac{P^n}{C^i} \times S_n$$

$$E_r^i = T_r^i \times C_f^i$$

where, C^i= concentration of contaminant 'i', C_n^i= abundance of non-contaminated samples (background value), T_r^i = toxicity coefficient, P_n= concentration of the specific polymer type of the MPs, Sn = hazard score of the MPs polymers.

The CMPI was calculated using the following formula:

$$\text{CMPI} = \frac{\text{Specific shape of MPs}}{\text{Total MPs}}$$

The CMPI was employed to quantify the relative environmental risks associated with different MPs shapes. This coefficient represents the proportion of a specific MPs shape (e.g., pellets, fibers) relative to the total number of MPs detected in a given sampling unit (Rangel-Buitrago et al. 2021). The CMPI is categorized into five classes, namely 0.0001–0.1 (minimum), 0.11–0.5 (average), 0.51–0.8 (maximum), and 0.81–1 (extreme).

Data analysis and statistics

The data is expressed as abundance of MPs/L in water, snow samples, and MPs/individual for fish samples. Further, descriptive statistics like mean and standard deviation (SD) were also calculated for each sample component, whereas the Shapiro-Wilk normality test was conducted for MPs abundance in surface water samples. We also employed Pearson correlation (95% significance) to test the relationship between different fish sample parameters (body weight, gut weight, total length, MPs abundance, and size). The p-values (two-tailed) less than 0.05 were considered as significant. The spatial distribution map (Fig. S1) was prepared in QGIS 3.40.1 using georeferenced sampling points, with color gradients applied to visualize spatial variations across the lake.

Results

MPs abundance and characteristics in abiotic (surface water, water column, snow) components

The MPs abundance with diverse shapes was ubiquitously observed in all the abiotic (water, snow) components of the lake (Table S1, Fig. 2). Across the 24 surface water sampling sites, a total of 2934 MPs were detected. The abundance of MPs was found to be normally distributed (Shapiro-Wilk test; $p = 0.58$). A large spatial variability in MPs abundance was observed in the surface water samples with a range of 59–188 MPs/L (Fig. 3a & Figure S1). The mean (± SD) abundance of MPs was 122 (± 29) MPs/L. For water column samples, depth-weighted concentrations were calculated by averaging MPs abundances at different depths, weighted by the portion of the water column each sample represented (Lenaker et al. 2019).

The mean depth-weighted MPs concentrations for the sites WC-1, 2 and 3 was 14, 9 and 8 MPs/L, respectively (Fig. 4a). The MPs abundance in snow samples varies between 11 and 18 MPs/L with a mean (± SD) of 14(± 2) for the eight sampling sites (Fig. 5a). A Kruskal–Wallis test revealed significant differences in MPs abundance among compartments ($\chi^2 = 37.14$, df = 3, $p < 0.001$).

We observed five major morphological types of MPs in the surface water, water-column, and snow samples. The morphological types detected were fiber, fragment, film, foam and pellet. The fiber (92.06%) was the most dominant type of MPs found in surface water samples (Fig. 3c) followed by fragment (6.15%), film (1.44%), foam (0.24%), and pellet (0.10%). The trend in dominance of MPs morphological types was similar in the water column sampling sites, with fibers (99.10%) being the most abundant, followed by fragment (0.90%) (Fig. 4c). In contrast to the surface and water-column samples, the snow samples

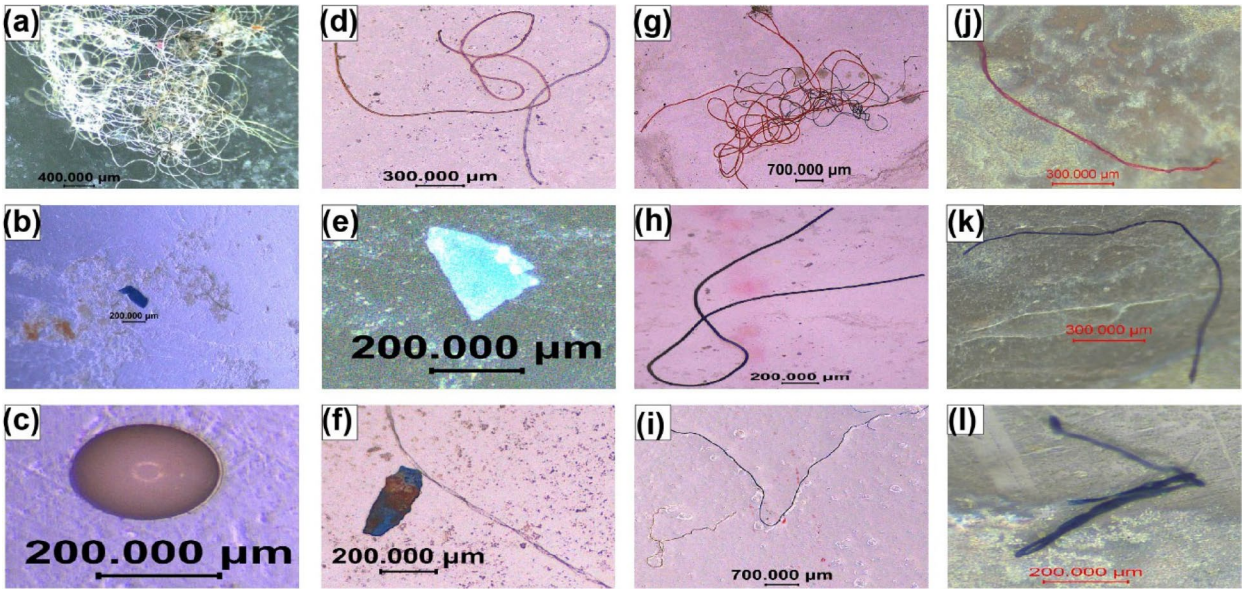

Fig. 2 The typical shapes of extracted MPs from (**a-c**) surface water, (**d-f**) water column, (**g-i**) snow and (**j-l**) fish samples of Manasbal Lake

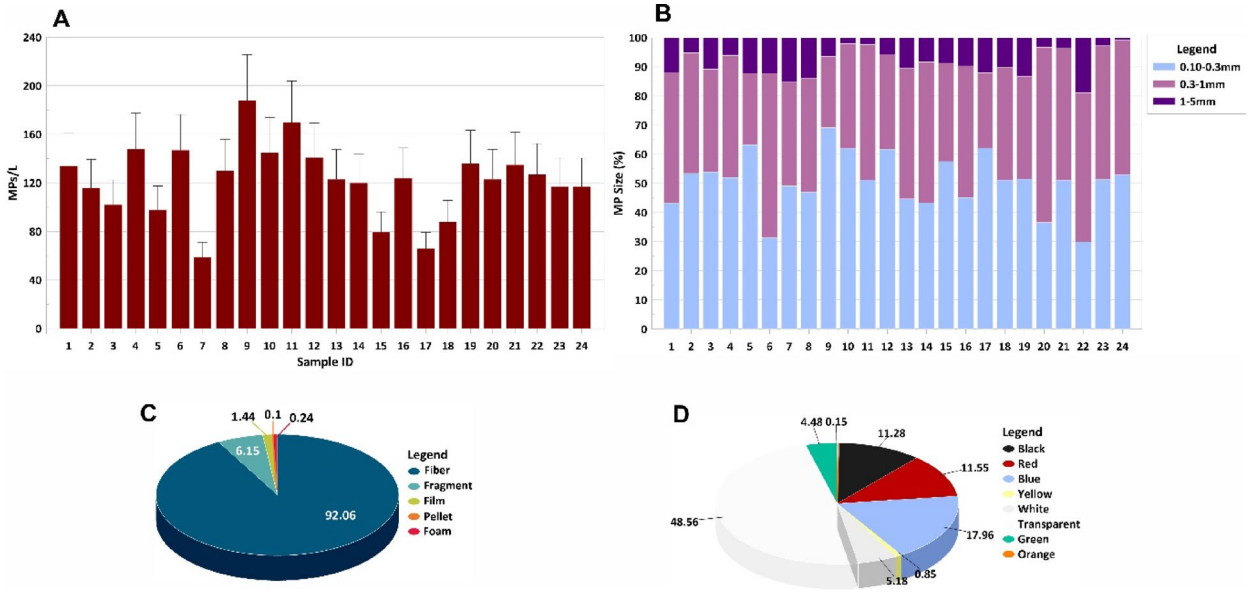

Fig. 3 The MPs abundance in surface water based on (**A**) sampling site, (**B**) size, (**C**) shape, and (**D**) color

were almost exclusively comprised of fibers (100%), with no other morphological types observed (Fig. 5c). The color distribution of MPs in surface water samples revealed the dominance of transparent particles (48.56%), followed by blue (17.96%) and red (11.55%) particles (Fig. 3d).

Other colors, including black (11.28%), white (5.18%), green (4.48%), yellow (0.85%) and orange (0.15%), were less abundant. For water column samples, a similar dominance pattern (Fig. 4d) of transparent (57.92%) MPs was observed, followed by blue (18.51%), black (12.80%), red (8.46%), green (2.10%), and white (0.21%). In snow samples, the MPs were exclusively composed of transparent (47.92%), black (31.07%), red (15.29%), and blue (5.72%) colors (Fig. 5d). A total of five different polymer types were identified in the surface water samples. Among these, 86.58% of the analysed MPs particles were PET, followed by high-density polyethylene (HDPE; 4.53%), polypropylene (PP; 1.51%), polyamide (PA; 4.44%), polystyrene (PS; 0.22%), while a 2.72% of the selected MPs could not be identified due to the lack of sufficient matching percentage (< 70%).

A similar polymer type pattern was observed for water column samples with PET (97.48%) being the most

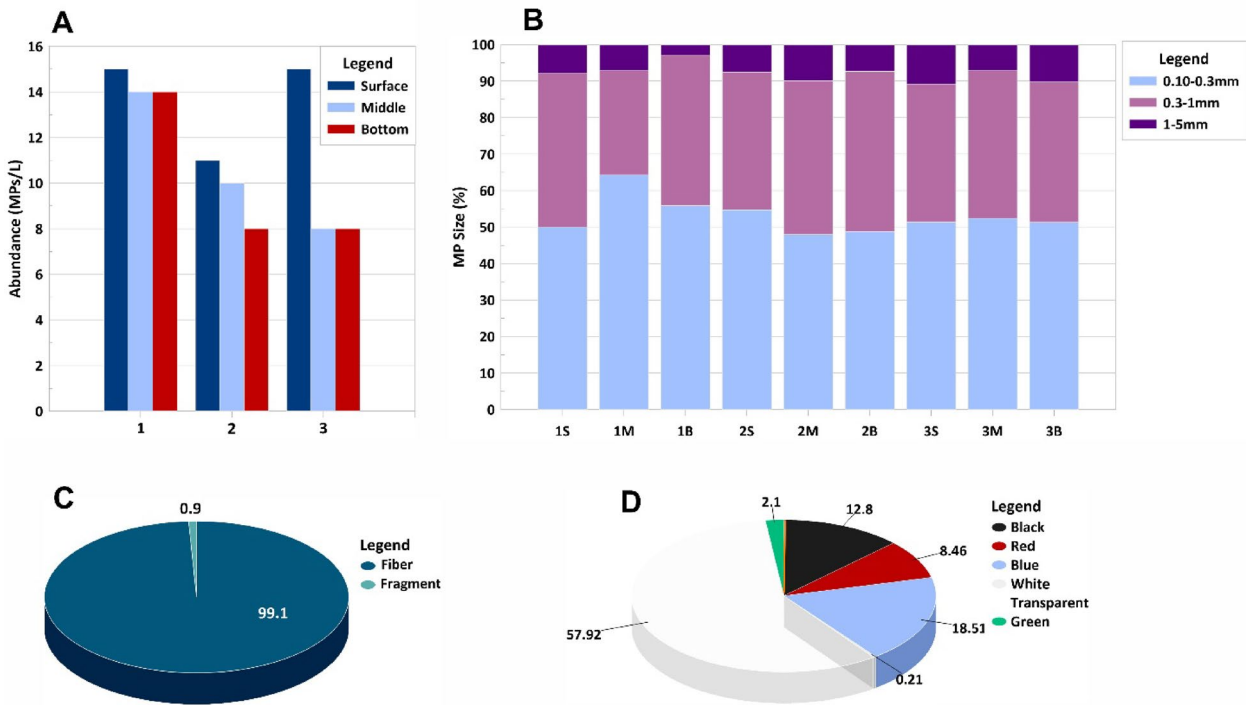

Fig. 4 The MPs abundace in water column samples based on (**A**) sampling site (**B**) size, (**C**) shape and (**D**) color. Capital letters S, M and B represents surface, middle and bottom sample from each site, respectively

abundant identified polymer type followed by HDPE, PP and others. Further, PET was the only polymer type observed for the snow samples.

MPs size in biotic and abiotic components

The results show that all three MPs size particles, viz. 0.1–0.3 mm, 0.3–1 mm and 1 to 5 mm are present in surface water (Fig. 3b), water column (Fig. 4b) and snow (Fig. 5b) samples. However, larger (1–5 mm) MPs were not observed in any of the fish samples (Fig. 6b) and three snow samples (MS-2, 4 and 5). The most dominant size range of MPs detected in surface water samples was 0.1–0.3 mm with 50.61%, followed by 0.3–1.3 (40.79%) and 1–5 mm (8.59%; Fig. 3b). The MPs abundance exceeded 100 items/L in most (~80%) of the surface sampling sites across the lake. The most dominant size fraction detected was 0.1–0.3 mm (52.96%), followed by 0.3–1.3 (39.14%) and 1–5 mm (7.89%; Fig. 4b). The one-way ANOVA results indicated no statistically significant difference in MPs abundance between depths (surface, middle, and bottom) ($p = 0.34$) or between sites (WC-1, WC-2, and WC-3) ($p = 0.10$). Similar to water samples, the MPs abundance in snow was dominated by the 0.1–0.3 mm size fraction (49.41%), followed by the 0.3–1 mm fraction (44.65%) and 1–5 mm fraction (5.94%; Fig. 5b).

In fish samples, MPs were primarily found in the 0.1–0.3 mm and 0.3–1 mm size fractions, with no detectable MPs in the 1–5 mm size fraction (Fig. 6b). The Pearson correlation analysis revealed several key relationships among the variables. Significant positive correlations (Fig. 8) were observed between total MPs and size categories (0.3–1 mm; $\rho = 0.97$; $p = < 0.001$ and 0.1–0.3.1.3 mm; $\rho = 0.94$; $p = < 0.001$), as well as between length and body weight ($\rho = 0.96$; $p = < 0.001$) and gut weight ($\rho = 0.88$; $p = < 0.001$).

MPs abundance and characteristics in the biotic (fish) component

The fish samples revealed a ubiquitous presence of MPs particles in their gastrointestinal tracts, irrespective of their body characteristics (length, weight). For the nine different fish samples, a total of 86 MPs particles were identified with a range of 7–13 MPs/individual and a mean(± SD) of 10(± 2); (Table S1; Fig. 6a). The morphological analysis revealed that all the detected MPs were fibers, with no other morphological types present (Fig. 6c). Regarding color distribution (Fig. 6d), transparent particles were the most dominant, accounting for 49.49%, followed by blue (24.25%), red (17.34%), and black (8.92%). Polymer analysis confirmed that all identified MPs were composed of PET (Fig. 7).

Discussion

MPs variability across biotic and abiotic components

The Manasbal Lake is one of the main tourist destinations with high human impacts. During fieldwork, we

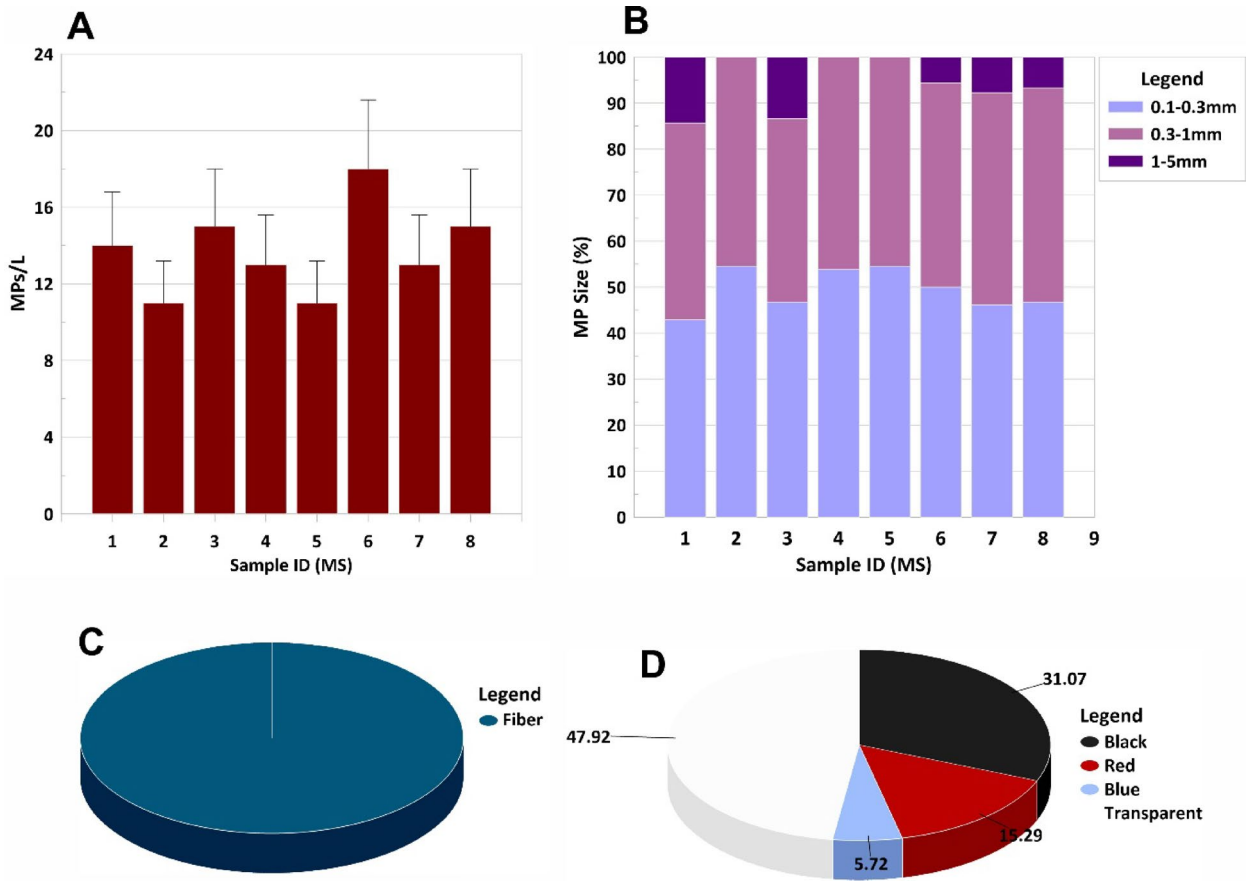

Fig. 5 The MPs abundance in snow samples based on (**A**) sampling site, (**B**) size, (**C**) shape, and (**D**) color

directly observed plastic bottles, shopping bags, food wrappers and plastic particles lying within the lake and along the shore possibly discarded by local population and tourists (Fig. 1d). A recent compliance report of the lake by the state pollution control committee (PCC) confirmed the dumping of unsegregated and untreated solid waste from domestic and commercial establishments at approximately 1 km from the Manasbal Lake (JKPCC 2024). The local government, however, launched several waste removal campaigns to mitigate the issue of anthropogenic pollutants in and around the Manasbal Lake (JKPCC 2024). Subsequently, MPs originated from catchment sources with direct linkages to anthropogenic activities. A previous sampling survey reported an average MPs abundance of 47 MPs/L in the surface water samples of Manasbal Lake (Kumar et al. 2025). Despite methodological differences, our data revealed a nearly threefold increase in MPs abundance, with an average of 122 MPs/L, indicating a significant rise in MPs abundance. This increased abundance may be attributed to enhanced local pollution sources, progressive fragmentation of existing plastic particles, and/or hydrological changes affecting retention in the lake. While recent mitigation efforts (JKPCC 2024) may have altered contamination levels, our data offers a critical pre-intervention reference for evaluating long-term trends.

Lake inlets, urban drainage, settlements, and roads in the catchment areas can directly or indirectly introduce MPs into the Manasbal Lake due to source proximity (Figure S1). We observed higher abundances of MPs at sites not directly adjacent to stream inlets, tourist parks, or residential areas, which suggests that lake hydrodynamics rather than source proximity controls the distribution and abundance of MPs in the Manasbal Lake (Figure S1). While residential areas and stream inlets are expected to be primary sources of MPs pollution, the relatively lower MPs abundances near these sites reveal efficient transport of MPs by surface water circulation to other parts of the lake (Chen et al. 2024). This interpretation is supported by our observation of the lowest spatial abundance of MPs near the lake inlets and built-up areas, mainly distributed through lake hydrodynamics (Kumar et al. 2025; Yan et al. 2021; Kowalski et al. 2016; Alfonso et al. 2021; Figure S1). Near the lake inlets, MPs may be more distributed to other parts of the lake due to higher flow. This spatial variability highlights the importance of considering both source proximity as well as the role of hydrodynamic processes while reporting MPs abundance

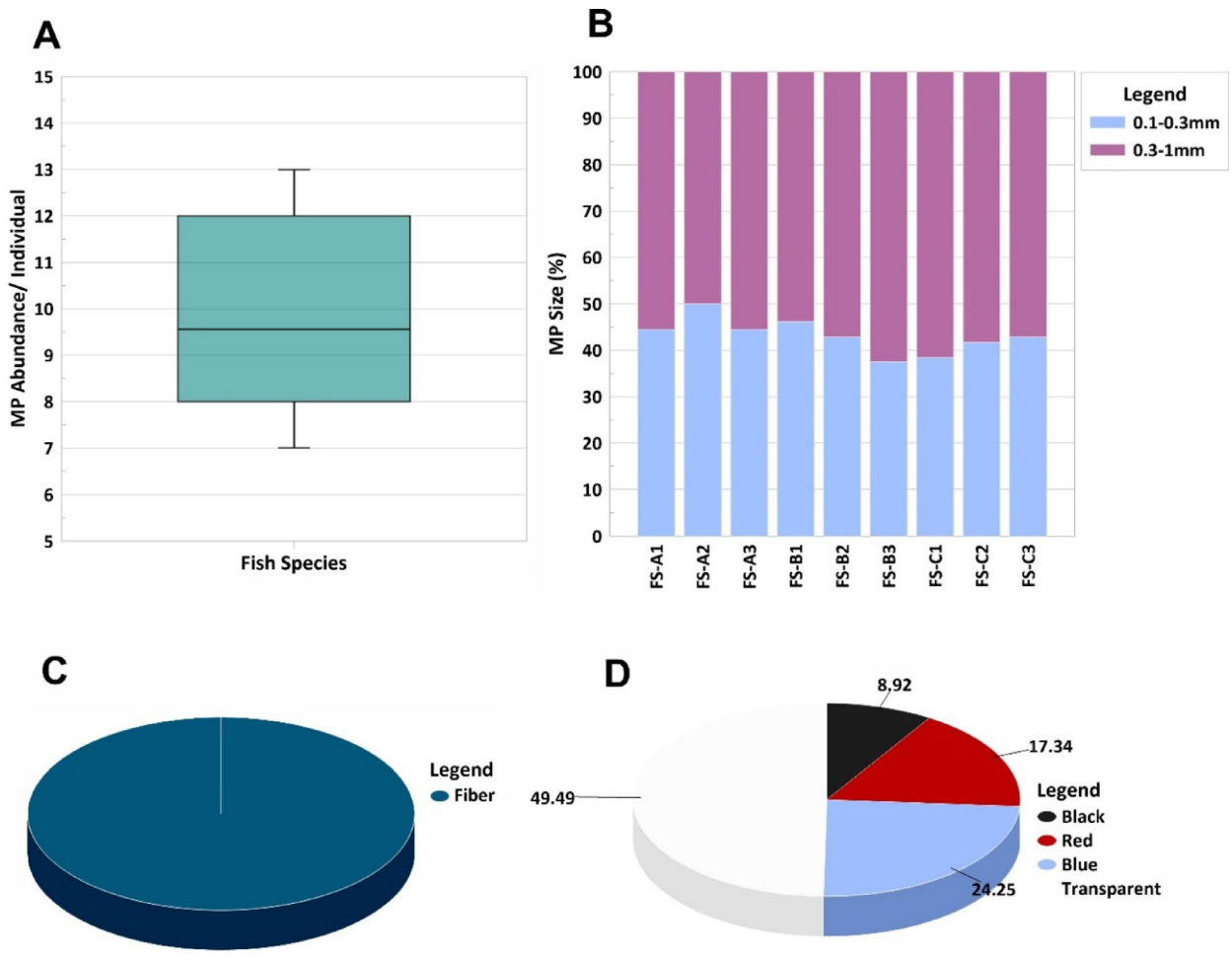

Fig. 6 The MPs abundance in fish samples based on (**A**) abundance/individual fish, (**B**) size, (**C**) shape, and (**D**) color

and their sources, particularly in open-type aquatic systems.

The abundance of MPs generally decreased with increasing size, being highest in the size class 0.1–1 mm and lowest in the class 1–5 mm. The dominance of the 0.1–0.3 mm size fraction, accounting for 50.61% of the total detected MPs, mirrors findings from similar studies where small MPs are predominant due to chemical and mechanical fragmentation of larger plastic particles (Leusch et al. 2023; Shi et al. 2023; Egessa et al. 2020; Welden and Lusher 2020). The MPs distribution observed in this study is comparable to patterns documented in other studies from the high-altitude lakes of the Himalaya (Mehboob et al. 2024). This size range (0.1–0.3 mm) is particularly significant because smaller particles enter the food web through direct ingestion by aquatic organisms (Setälä et al. 2014).

Atmospheric flux and variability of MPs in the water column

The presence of MPs in snow samples is indicative of atmospheric deposition, a mechanism well-documented for transporting MPs to remote regions, including polar and mountainous areas (Mehboob et al. 2024; Wen et al. 2024; Pastorino et al. 2022; Sun et al. 2022), including the Himalaya (Ankit et al. 2024; Mehboob et al. 2024). Specifically, westerly winds common in the western Himalaya are considered as potential atmospheric source of long-range transport for MPs (Ankit et al. 2024). Snowfall in high-altitude areas of Kashmir Valley occurs during the winter months (December-March) when the westerly winds are most active (Lone et al. 2020). Additionally, these winds can enhance the trapping and accumulation of airborne MPs pollutants, which are then directly deposited on the surface or released as the snow melts, entering aquatic systems. Further research is needed to understand the significant role of long-range atmospheric MPs inputs to high-mountain lakes of the Indian Himalaya. However, such an analysis is beyond the scope of our current work. Further, given the similar patterns observed across different matrices in our study, we acknowledge that atmospheric influence may be relatively minor, with direct sources playing a more significant role in delivering MPs. Our study shows an average

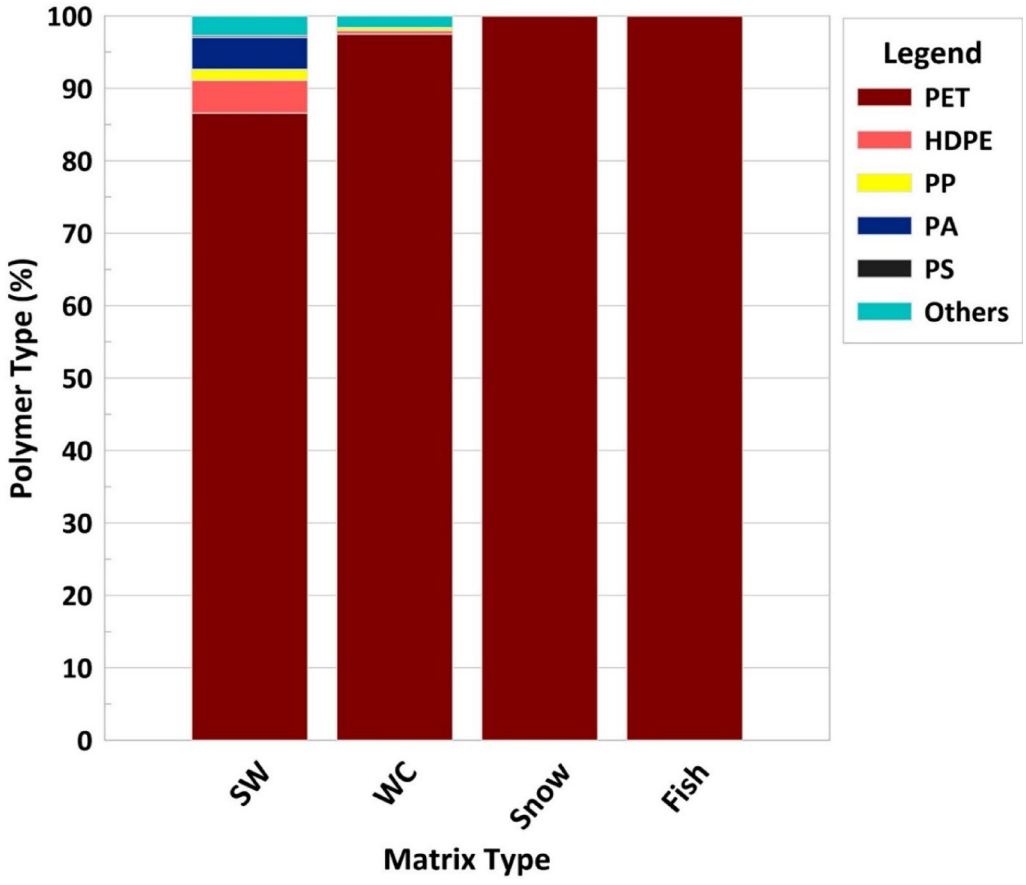

Fig. 7 The percentage abundance of different polymer types in each biotic and abiotic component type of the lake

abundance of 14 MPs/L in the snow samples. This highlights a minor proportion of atmospheric pathways in delivering MPs to the lake, though further quantitative research is needed to account for potential losses or redistribution processes.

The lack of significant differences in MPs abundance between surface, middle, and bottom water layers ($p = 0.341$) suggests that MPs in the lake are uniformly distributed throughout the water column. This finding is consistent with studies in river systems and lakes where MPs were found to be evenly dispersed due to the hydrologic environment, local water mixing, and polymer density (Pasquier et al. 2023; Lenaker et al. 2019). However, a relatively higher abundance of MPs near the surface of the water column could be due to the buoyancy and slower vertical mixing. The relatively deep and well-mixed nature of Manasbal Lake could explain the homogeneous distribution of MPs observed in this study. The higher MPs concentration in lake surface water compared to catchment snow suggests that local pollution sources, such as wastewater discharge, tourism, and surface runoff, are the primary contributors rather than long-range atmospheric deposition. Hence, localized direct point and non-point inputs and local human activities seem to play a significant role in MPs contamination of the Manasbal Lake.

Morphological and color characterization of MPs

The morphological analysis of MPs revealed that fibers were the most represented type, accounting for almost 90–100% of particles in all compartments due to their low density and physical characteristics in possibly all environmental compartments (Aves et al. 2022; Bullard et al. 2021). This aligns with previous findings, where fibers, particularly those from textiles, cage fishing, cloth washing near banks, and ropes used in boats have been identified as the most prevalent form of MPs (Rahman et al. 2025; Rodrigues et al. 2018a, b; Browne et al. 2011). The human activities (washing clothes, fishing) and waste dumping of plastic provide MPs with sources of different types and sizes once disintegrated (Malla-Pradhan et al. 2022). The dominance of fibers is also consistent with studies from lakes in the Indian subcontinent (Resmi and Vaishnavi 2025; Neelavannan et al. 2022; Ajay et al. 2021). Synthetic textiles are one of the largest sources of MPs due to the widespread use of synthetic fibers in clothing and other consumer products. The presence of fibers across all sampling compartments suggests that these

particles are not only pervasive in the lake but also easily transported and deposited through various environmental mechanisms (Hossain and Engelhardt 2025). Hence, it should be considered that fibers found in high-altitude lake environments are not always of anthropogenic origin. Atmospheric deposition has also been identified as a source of fiber-type MPs from various origins (González-Pleiter et al. 2020).

The color distribution of MPs also provides insight into potential sources. Transparent particles were dominant in all sampling compartments and accounted for ~50% of the total MPs count, which could be indicative of the widespread use of transparent plastics in consumer goods and packaging (Lusher et al. 2020). The high abundance of transparent MPs, despite the dominance of colorful plastic products in the market, suggests that transparent MPs may originate from the breakdown of larger plastic materials like plastic bags, fishing lines, and packaging films, which fragment over time (Sun et al. 2022; Egessa et al. 2020). Previous studies have also confirmed that the use of acids for organic matter removal might affect MPs color due to the bleaching process (Aslam et al. 2022). Further, blue, black, and red particles frequently observed in all samples also suggest potential sources from textile fibers, as these colors are common in synthetic fabrics.

The predominance of transparent, blue, and red colors in fish gastrointestinal tracts suggests selective ingestion of frequently available and certain colored MPs, a phenomenon previously reported in other aquatic organisms (Okamoto et al. 2022). The detection of MPs in the gastrointestinal tracts of fish provides clear evidence of bioaccumulation of MPs within the lake's food web (Horie et al. 2024). The mean abundance of 10 ± 2 MPs/individual, with fibers being the sole morphological type detected, indicates that fish in Manasbal Lake are consuming MPs, through ingestion of contaminated water or prey. The fact that only smaller MPs (0.1–1 mm) were found in fish samples, with no MPs > 1 mm, aligns with findings from other studies, where smaller MPs are more easily ingested by aquatic organisms (Roch et al. 2020). This trend is of concern because smaller MPs being more abundant will pass through biological barriers easily and enter the food chain, with potential long-term ecological and health implications for aquatic species. A significant positive correlation between MPs abundance and smaller MPs fractions (0.1–0.3 mm & 0.3–1 mm) indicates that finer MPs are more bioavailable, making ingestion more dependent on particle size rather than individual fish characteristics (Fig. 8). This aligns with recent studies suggesting that MPs retention is governed by feeding behaviour, gut passage dynamics, and environmental MPs availability rather than body size alone (Dawson et al. 2025; Li et al. 2021; de Vries et al. 2020).

Polymer characterisation and its variability in biotic and abiotic compartments

Polymer analysis confirmed that PET was the dominant polymer type in all environmental matrices, including surface water, water column, snow, and fish samples, constituting over 80–100% of the identified MPs particles (Fig. 7 & Figure S2). The polymer type reflects the particle density and their behaviour in aquatic systems. For instance, depending on their density, MPs particles may float or sink to the deeper water column when discharged in aquatic systems. However, it is now highlighted that apart from density, water temperature, water currents, wind, salinity, biofouling, as well as changes of surface to volume ratio also affect the vertical distribution of MPs (Egessa et al. 2020 and references therein). Hence, the high abundance of PET in surface water samples, despite its high density, can be attributed to its fragmentation into smaller, buoyant particles, wind-generated waves keeping it suspended, and biofouling enhancing its buoyancy. Additionally, continuous anthropogenic inputs might also contribute to its prevalence in both biotic and abiotic components. PET is widely used in products such as plastic bottles, textiles, and packaging materials, with almost 60% of all PET produced for synthetic fibres and 30% for plastic bottles (Aves et al. 2022 and references therein). Moreover, recent studies highlighted that among all polymers, PET is seen as a major polymer type (39%), followed by PE (30%), PP (19%), and PS (< 1%) in freshwater ecosystems of India (Pandey et al. 2023). These sources subsequently become major contributors to MPs pollution in aquatic environments (Sarkar et al. 2021). The near-complete dominance of PET in the snow samples indicates that atmospheric deposition may be a significant pathway for the distribution of this polymer to remote regions, supporting findings from similar studies in high-altitude lakes (Aves et al. 2022; Parolini et al. 2021a, b).

The polymer analysis confirms that all MPs in fish were composed of PET, which further supports the hypothesis that human-made plastics, particularly those from packaging and textiles, are the primary sources of contamination in Manasbal Lake. This finding mirrors results from similar studies in fishes of the Indian subcontinent (Devi et al. 2024; Pandey et al. 2023). Given the widespread use of PET-based products in urban and rural areas surrounding the lake, it is suggested that local anthropogenic activities, including tourism, fishing, and plastic waste mismanagement, contribute significantly to MPs pollution in both the lake's water and its biota. The results are also in coherence with the polymer composition of MPs reported from several other studies from Asia, where PET and PP dominate the environmental matrices (Pandey et al. 2023; Piyawardhana et al. 2022).

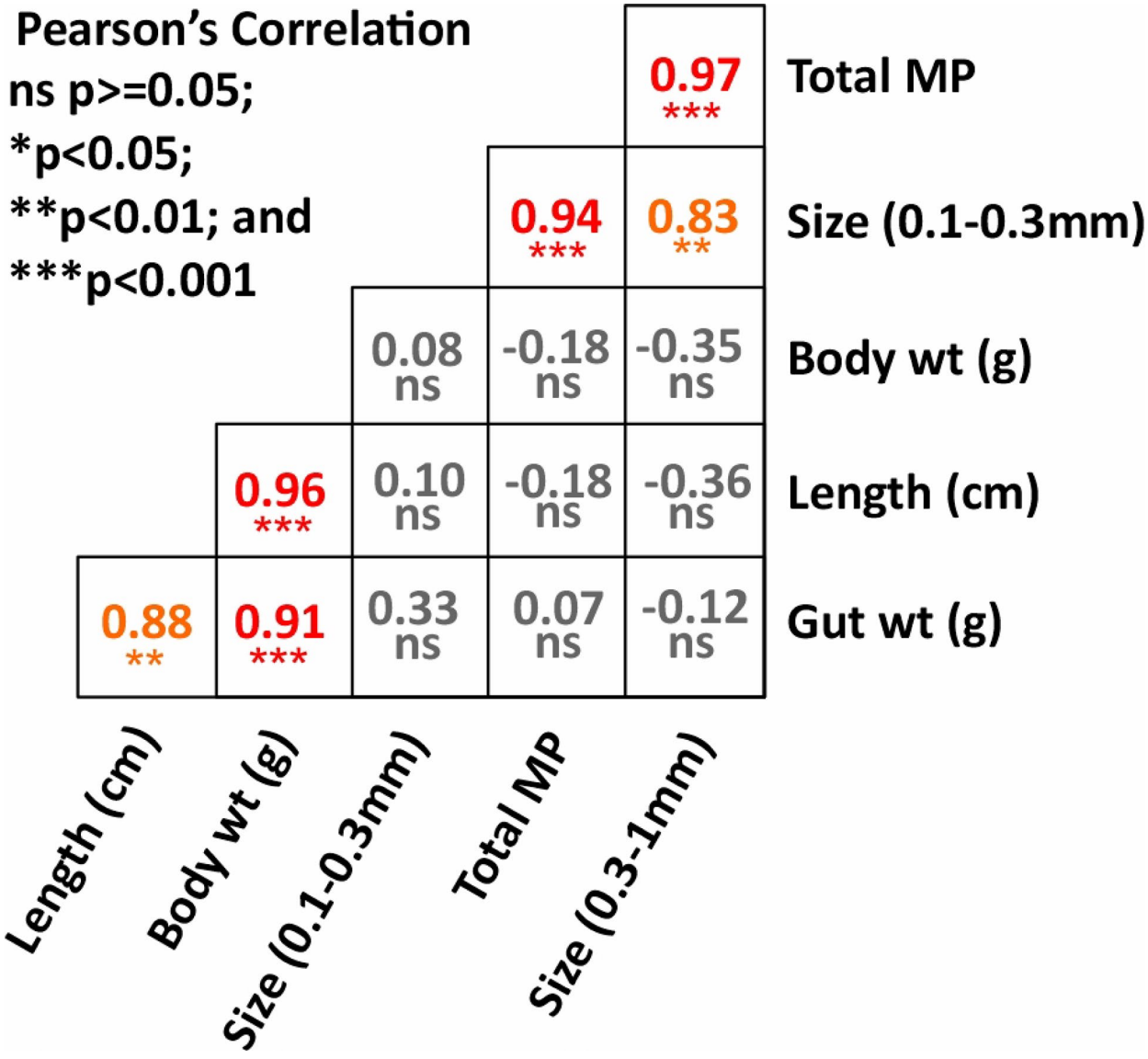

Fig. 8 The correlation analysis between fish characteristics (length, body weight, gut weight, MPs size) and abundance of MPs (MPs/individual). ns (not significant)

Risk assessment

The risk assessment of MPs in environmental components helps evaluate the potential hazards they pose to ecosystems and human health. In this study, the PHI and PERI indices were used to assess the impact of MPs in different biotic and abiotic components of Manasbal Lake, including surface water, water column, snow, and fish (Table 1). The results showed that surface and water column samples had very high hazard levels (PHI = 815.17 and 225.67), with extreme toxicity mainly driven by MPs fibers as per CMPI.

Snow and fish exhibited hazard levels of III (PHI = 55 and 38.22), with fibers still being the most significant contributor to impact. These findings highlight that MPs contamination, particularly fibers, poses a substantial risk to the lake's ecosystem, with the most severe effects observed in the water compartment. The PERI indicated that surface water has an extreme toxicity level, posing the highest environmental risk. Both snow and fish samples exhibited a high toxicity level, while water column samples were classified as having a dangerous toxicity level. The CMPI analysis revealed that fibers have an extreme impact across both biotic and abiotic components.

MPs pollution in other Western Himalayan lakes

The findings from this study, when compared with previous reports, reveal consistent trends underscoring the pervasiveness of MP contamination across western Himalayan Lake ecosystems. Although our methodology

Table 1 Values of risk assessment indices in PHI, PERI and CMPI for biotic and abiotic components in manasbal lake

Component	PHI	Hazard	PERI	Toxicity level	CMPI (Fiber)	Impact	CMPI (Fragment)	Impact	CMPI (Film)	Impact	CMPI (Pellet)	Impact	CMPI (Foam)	Impact
Surface water	815.17	IV	1381.64	Extreme	0.92	Extreme	0.06	Minimum	0.01	Minimum	0	Minimum	0	Minimum
Water column	225.67	IV	578.63	Danger	0.99	Extreme	0.01	Minimum	0	Minimum	0	Minimum	0	Minimum
Snow	55	III	500	High	1	Extreme	0	Minimum	0	Minimum	0	Minimum	0	Minimum
Fish	38.22	III	546.03	High	1	Extreme	0	Minimum	0	Minimum	0	Minimum	0	Minimum

excluded particles < 100 µm, limiting direct quantitative comparison with studies using smaller sieve sizes (e.g., 0.02 mm). However, qualitative patterns such as dominant polymer types, color, and spatial variability remain comparable and highlight common anthropogenic influences, notably tourism and waste discharge.

Recent investigations across the Indian Himalaya confirm the ubiquity of MPs in water and sediments, emphasizing anthropogenic inputs and atmospheric deposition as major contributors (Talukdar et al. 2023). Backward air-trajectory analyses and precipitation records further demonstrate that fine MPs can be transported long distances from western and urban regions and deposited through wind, rain, and snowfall (Ankit et al. 2024). Such mechanisms enhance MP accumulation even in remote, high-altitude sites. Studies from Ladakh lakes (Tsomoriri, Tsokar, and Pangong) reported abundant fine MPs dominated by transparent and white colors, suggesting inputs from atmospheric fallout and fragmentation of larger plastics in minimally disturbed regions (Tsering et al. 2022). In contrast, lowland lakes such as Anchar, Dal, and Manasbal experience heavier anthropogenic pressures from urban runoff, tourism, and domestic waste. High fiber abundance and dominance of white MPs (Rather et al. 2025; Neelavannan et al. 2022; Nazir et al. 2024a, b) corroborate these localized sources, while habits such as packaging and cosmetic use further contribute to pollution (Nazir et al. 2024a, b). Similar contamination levels have been reported in Renuka (Ajay et al. 2021), Rewalsar (Bulbul et al. 2023), and Nainital Lakes (Jain et al. 2024), where tourism and wastewater inputs serve as key drivers. Kumar et al. (2025) further showed that MP distribution is linked to domestic sewage and sediment grain size in Manasbal Lake.

Collectively, these studies demonstrate that Himalayan Lakes function as dynamic sinks for MPs, influenced by both catchment hydrodynamics and long-range atmospheric deposition. The convergence of local human activities (tourism, fishing, poor waste management) with regional atmospheric inputs accentuates the persistence of MPs in these fragile environments. This synthesis highlights the need for integrated management, improved waste handling, and expanded research on MPs pathways and ecological risks in high-mountain freshwater systems of the western Himalaya.

Possible limitations and future directions

While this study aims to provide a detailed discussion of the MPs contamination in biotic and abiotic components of the lake, we acknowledge the limitation of having conducted sampling only once and a limited number of fish ($n = 9$) and snow samples ($n = 8$). Further, our methodology excluded particles < 100 µm, potentially underestimating MPs loads and risks, thereby restricting direct quantitative comparisons with studies using smaller sieve sizes. The relatively modest sample sizes for both fish and snow limited the ability to draw robust statistical conclusions, particularly in evaluating trophic transfer or bioaccumulation of MPs. While snow sampling at two sites offered valuable insight into winter atmospheric deposition patterns, a more comprehensive, multi-seasonal monitoring program is needed to fully capture the temporal variability and magnitude of atmospheric flux.

Despite sampling limitations, this study provides a systematic baseline report of MPs pollution in the biotic and abiotic components of the Manasbal Lake, providing critical insights into legacy contamination and spatial distribution patterns as well as atmospheric fluxes. Future studies concerning MPs pollution in biotic and abiotic lake components should also prioritize increased biological sampling across trophic levels, and broader spatial coverage of snow and water sampling from this region to enhance statistical power and ecological relevance of the results. Given the vulnerability of high-altitude lakes, dedicated monitoring of infrastructure is urgently needed to track MPs fluxes in this fragile region.

Conclusions

In this study, MPs were assessed in both biotic and abiotic compartments of the Manasbal Lake. MPs were observed in all compartments, with fibers representing a high proportion across the samples. The findings further reveal that MPs abundance is lowest near the proximity sources and inlets and highest near the outlet of

Manasbal Lake. Our results reveal that localized sources, such as wastewater discharge, tourism, fishing, and surface runoff, are the primary contributors of MPs pollution rather than long-range atmospheric deposition in the Manasbal Lake. Risk assessment reveals that fibers possess a substantial risk to the Manasbal Lake's ecosystem, with the most severe effects observed in the surface water. It is recommended to raise awareness among the local population and tourists to manage MPs pollution and promote Manasbal Lake as an eco-friendly destination. In addition, continuous monitoring and integrated waste management are essential to reduce MPs inputs and safeguard the ecological integrity of Himalayan freshwater systems, providing critical guidance for future policy and ecosystem management strategies.

Abbreviations

ATR	Attenuated Total Reflectance
CMPI	Coefficient of microplastic impact
FTIR	Fourier Transform Infrared Spectroscopy
HDPE	High-density polyethylene
IAEC	Institute Animal Ethics Committee
ISM	Indian Summer Monsoon
MPs	Microplastics
PET	Polyethylene terephthalate
PCC	Pollution control committee
PHI	Polymer hazard index
PERI	Potential ecological risk index
QA	Quality assurance
QC	Quality control
SD	Standard deviation
$ZnCl_2$	Zinc chloride

Supplementary Information

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Supplementary Material 1

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Author contributions

Aasif Mohmad Lone : Writing –original draft, Writing – review & editing, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Keshava Balakrishna** : Supervision, Funding acquisition, review & editing. **Shujith Balaji** : Formal analyses, review & editing. **Haroon Yehya** : Formal analyses, review & editing. **Gokul Valsan** : Formal analyses, review & editing. **Anjali Tamrakar** : Formal analyses, review & editing. **Anish Kumar Warrier** : Supervision, Resources, review & editing.

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Data availability

The datasets analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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